

**CO900G – ENVIRONMENTAL SCIENCE AND
SUSTAINABLE ENGINEERING**

INDIVIDUAL TECHNICAL REPORT

Environmental Problem: Traffic Congestion and Vehicle Emissions

**Proposed Engineering/IT Solution: Vehicle Emission Estimation &
Monitoring System-Making Users Aware of Their Transportation
Emissions**

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1. ABSTRACT

Traffic congestion and vehicle emissions have become major environmental challenges in urban and semi-urban areas. Large numbers of vehicles operating on roads increase fuel consumption and contribute to emissions such as carbon dioxide (CO₂), carbon monoxide (CO), nitrogen oxides (NO_x), hydrocarbons and particulate matter. These emissions affect air quality, contribute to climate change and may create risks to human health.

This report proposes a Vehicle Emission Estimation and Monitoring System designed to improve awareness of transportation-related emissions. The proposed system estimates emissions using information such as distance travelled, fuel type, vehicle category and an appropriate emission factor. Instead of directly measuring exhaust gases, the system provides an accessible estimate that helps users understand the environmental impact of their travel choices.

The solution is examined using sustainable transportation, the United Nations Sustainable Development Goals (SDGs), ISO 14001 environmental management, Life Cycle Assessment (LCA) and Environmental Impact Assessment (EIA). Technical feasibility, environmental benefits, limitations and future improvements are also considered.

2. INTRODUCTION

Transportation is essential for modern society because it enables people and goods to move between homes, educational institutions, workplaces, commercial areas and other locations. However, the rapid growth in the number of vehicles has also increased traffic congestion and fuel consumption. In many urban and semi-urban areas, congestion results in longer travel times, frequent stopping and starting, idling and inefficient fuel use.

Vehicle emissions are an important environmental consequence of road transportation. Fuel combustion releases greenhouse gases and air pollutants into the atmosphere.

Carbon dioxide is a major greenhouse gas associated with fossil-fuel consumption, while pollutants such as nitrogen oxides, carbon monoxide, hydrocarbons and particulate matter can affect local air quality.

An important part of addressing this problem is improving public awareness. People may understand that vehicles produce emissions, but they may not know approximately how much their individual journeys contribute. A simple emission estimation and monitoring system can convert everyday travel information into understandable environmental information. This can help users compare transportation choices and encourage lower-emission behaviour.

The proposed project focuses on an awareness-oriented digital solution rather than attempting to directly control traffic or physically measure exhaust gases.

3. PROBLEM STATEMENT

Increasing vehicle usage and traffic congestion contribute to higher fuel consumption and transportation-related emissions. In congested conditions, vehicles may spend considerable time idling or moving at low speeds, which can increase fuel consumption compared with smoother travel.

A major difficulty is that individual users generally do not have a simple way to understand the environmental impact of their journeys. Without accessible information, it can be difficult for people to compare different travel choices or recognize how repeated vehicle use contributes to emissions.

The problem addressed by this project is therefore the lack of an accessible system for estimating and monitoring the transportation emissions associated with individual journeys.

4. SUSTAINABILITY ANALYSIS

Sustainable transportation aims to satisfy mobility needs while reducing negative environmental and social impacts. The proposed system supports this mainly through

awareness. By showing estimated emissions for journeys, it can encourage users to consider alternatives such as public transportation, walking, cycling, carpooling or combining multiple trips. It can also encourage efficient driving and reduced unnecessary travel.

The solution is relevant to SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action).

ISO 14001 provides a framework for environmental management systems. Although this software project is not itself an ISO 14001-certified environmental management system, its development can be considered through Plan–Do–Check–Act: identify the problem and objectives, develop and operate the system, check results and performance, and improve the system continuously.

Life Cycle Assessment (LCA) concepts can be applied by considering electricity use, computing resources, storage, network use, device operation, maintenance and hardware replacement. Environmental Impact Assessment (EIA) is generally associated with physical projects; however, EIA principles can be used conceptually to consider both the positive awareness benefits and the environmental impacts of the digital infrastructure.

5. PROPOSED ENGINEERING/IT SOLUTION

The proposed Vehicle Emission Estimation and Monitoring System is an information-technology-based solution that estimates transportation emissions from user-provided travel information. A typical user would enter information such as travel distance, vehicle type, fuel type and, where appropriate, fuel consumption. The system would process these inputs using an emission factor and calculate an estimated quantity of CO₂ produced by the journey. Results can be recorded so users can monitor their travel emissions over time.

The solution is primarily an awareness and monitoring system. It should not be interpreted as a replacement for laboratory emission testing or direct exhaust-gas measurement.

6. SYSTEM WORKING AND EMISSION ESTIMATION METHODOLOGY

The basic process is:

1. The user enters travel information.
2. The system identifies the relevant vehicle and fuel category.
3. An appropriate emission factor is selected.
4. The estimated emission is calculated.
5. The result is presented.
6. Previous results can be monitored to identify trends.

A simple approach is:

Estimated CO₂ Emissions = Distance Travelled × Emission Factor

Alternatively, where reliable fuel-consumption data are available:

Estimated CO₂ Emissions = Fuel Consumed × Fuel Emission Factor

The accuracy depends on the quality and suitability of the emission factor and the information supplied. Vehicle type, fuel type, efficiency, traffic congestion, idling, driving behaviour, maintenance, road conditions and weather can affect emissions. Therefore, results should be clearly identified as estimates.

7. KEY FEATURES OF THE PROPOSED SYSTEM

The proposed system can include journey-based emission estimation, support for different vehicle and fuel categories, emission history, comparison of journeys or transportation choices, simple environmental indicators, general recommendations for reducing transportation emissions, updated emission factors as better data become available, and input validation to reduce errors.

8. FEASIBILITY ANALYSIS

Technical feasibility is high because the core calculation uses common programming concepts such as input validation, arithmetic operations, data storage and user interfaces. A basic version does not require specialized laboratory equipment.

Economic feasibility is also favourable because the basic system can be developed using standard computing resources and commonly available software technologies. Costs would mainly involve development, hosting, maintenance, data acquisition and optional third-party services.

Operationally, the system is feasible because users can provide basic travel information without specialized technical knowledge. A simple interface and clear explanation of assumptions are important for user acceptance.

9. EXPECTED ENVIRONMENTAL BENEFITS

The direct environmental benefit comes from improved information rather than physical emission removal. Potential benefits include increased awareness of transportation-related CO₂ emissions; encouragement of public transport, walking, cycling and carpooling where practical; awareness of unnecessary trips and inefficient travel patterns; encouragement of fuel-efficient driving; and better understanding of the relationship between travel behaviour and environmental impact.

Actual emission reductions would depend on whether users change their transportation behaviour after receiving the information.

10. JUSTIFICATION OF THE PROPOSED SOLUTION

The proposed system is justified because it addresses an important environmental issue using an accessible engineering and IT approach. Although a software awareness system cannot solve every aspect of traffic congestion and emissions, awareness can support behavioral change.

The project is appropriate for an environmental engineering and computer science context because it combines environmental science with information technology. It demonstrates how a computational model can communicate environmental impacts and support sustainable decision-making while applying concepts such as SDG's, environmental management, life-cycle thinking and impact assessment.

11. EXPECTED CONTRIBUTION TO SUSTAINABILITY

The project is expected to strengthen the connection between individual transportation choices and environmental consequences. When users can see estimated emissions associated with journeys, environmental impact becomes more tangible and easier to compare.

The system supports sustainability by promoting awareness, encouraging responsible transportation choices and demonstrating how digital technologies can support environmental objectives. Its long-term value depends on user adoption, reliability of estimates and measurable changes in travel behaviour.

12. CONCLUSION

Traffic congestion and vehicle emissions are significant environmental challenges associated with increasing road transportation. They contribute to greenhouse gas emissions and air pollution while also increasing fuel consumption and reducing transportation efficiency.

The proposed Vehicle Emission Estimation and Monitoring System provides an IT-based approach for improving awareness of transportation-related emissions. By using travel and vehicle information with suitable emission factors, the system can estimate CO₂ associated with individual journeys and help users monitor their transportation footprint.

The project is technically and economically feasible at a basic level and can be expanded with improved data sources and more advanced estimation methods. Its main contribution

is not direct emission control, but accessible information that can encourage sustainable transportation behaviour.

13. REFERENCES

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